

BUILDER DESIGN PATTERN

Creational Design Pattern

Builder Design Pattern

When creating an object that needs many **different settings** – especially when most of them are optional – the constructor becomes messy and confusing.

This creates several problems:

1. Extremely long parameter lists in the constructor
2. Hard to remember which parameters are required and which are optional
3. No flexibility to set only specific values you need

Builder Design Pattern

Problem: When a class constructor needs too many parameters (especially optional ones), the Builder Pattern lets you construct complex objects step-by-step in a clear and readable way.

Solution: Splits the object creation process into separate steps using a builder class. You can set only the properties you need using a fluent interface (method chaining), making the code clean and easy to understand.